Pratyaksh Tomar

+91-8287642034 Pratyakshtomar2000@gmail.com Linkedin Github Code360 Leetcode Portfolio

Career Objective

Aspiring Software Engineer with a strong foundation in data structures, algorithms, and object-oriented programming. Proficient in C++, Python, and TypeScript, with project experience in AI agents, machine learning, and network systems. Seeking an entry-level Software Engineer position to apply analytical and problem-solving skills in building reliable, efficient software while growing through real-world engineering challenges.

Skills

Programming: C, C++, Python, JavaScript, TypeScript

Frontend: HTML5, CSS3, Tailwind CSS

Backend: FastAPI, Flask, Bun, REST APIs

AI/ML: Pandas, Matplotlib, Scikit-learn

GenAI: OpenRouter API, Groq API, LangChain, RAG

Databases: MongoDB, MySQL

Tools: Git, GitHub, Vercel

Core CS: DSA, OOP, DBMS, Computer Networks, OS

PROJECTS

ClawNest | *TypeScript, Bun, OpenRouter API* **Live** | **Github**

- Built an **open-source AI agent** in **TypeScript** on **Bun**, usable via a terminal CLI and **Telegram bot**.
- Engineered an **autonomous agent loop** with tools for files, shell, code search and web scraping.
- Integrated **OpenRouter** to switch between multiple LLMs, with **Agent, Plan and Ask** modes.
- Designed a **staged approval workflow** with git-style diff previews and owner-only Telegram access.

California Housing Predictor | *Python, Flask, scikit-learn, Chart.js, Pandas* **Live** | **Github**

- Built a full-stack **ML web app** with **Flask** and **scikit-learn** to predict California house values.
- Trained a **Random Forest** achieving **R^2 of 0.83** and **MAE of $30.9K** on 20,640 records.
- Engineered a **preprocessing pipeline** with imputation, scaling and encoding, using a stratified 80/20 split.
- Developed a **REST API** with batch CSV inference and a **Chart.js** evaluation dashboard.

DPI Engine | *C++17, Multithreading, TCP/IP, TLS, PCAP* **Github**

- Built a **Deep Packet Inspection engine** in C++17 to parse PCAP traffic and classify apps by 5-tuple.
- Extracted **SNI from TLS Client Hello** and **HTTP Host headers** to block traffic by IP, app and domain.
- Designed a **multithreaded pipeline** with consistent hashing and thread-safe queues for fast processing.
- Added a **CLI** with reporting features to review classified applications and blocked traffic.

Education

Dr. A.P.J. Abdul Kalam Technical University, Lucknow **2023 – 2027**
Bachelor of Technology in Computer Science Engineering

Dehradun Public School, Ghaziabad **2019 – 2020**
Secondary X – CBSE

Dehradun Public School, Ghaziabad **2021 – 2022**
Senior Secondary XII – CBSE

ACHIEVEMENTS

- Selected as a **finalist** in **Policy Conclave '26** at **Indian Institute of Technology Kanpur**.
- Competed in the **National AI/ML Hackathon** by **AVEVA** (organised by **IIT Hyderabad**), building AI/ML solutions as part of team Thunders, Mar 2026.
- Participated in **Hacker's Premier League** at **Guru Gobind Singh Indraprastha University, Delhi**.
- Awarded Top Performer certificate in **E&ICT IITG: Introduction to Programming** by **Coding Ninjas × IIT Guwahati**, Feb 2025 (NSDC certified).
- Solved **400+ coding problems** across platforms including **LeetCode** and **Code 360**.